

# MOCK AMPLIFIER DESIGN

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Course: EE210

July 3, 2026

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## Abstract

This report documents the complete design, theoretical analysis, and Cadence Virtuoso simulation of an analog Common-Source amplifier engineered to drive a  $1\text{ k}\Omega$  AC-coupled load at an operating frequency of 100 MHz. The primary objective was to achieve a fundamental-to-third-harmonic distortion ratio (HD1/HD3) of at least 40 dB for a 1 V peak-to-peak output swing, operating from a single 1.8 V supply without the use of inductors. Through careful device sizing ( $W = 20\mu\text{m}$ ,  $L = 500\text{nm}$ ) and biasing ( $V_{GS} = 700\text{ mV}$ ), the amplifier successfully achieved a peak-to-peak swing of approximately 1 V. Spectral analysis via Discrete Fourier Transform (DFT) yielded an HD1/HD3 ratio of 31.89 dB. The deviation from the 40 dB target is mathematically analyzed and attributed to the transient settling time of the AC-coupling network relative to the simulation window.

## 1 Introduction and Design Objective

The development of highly linear analog amplifiers at high frequencies is a fundamental challenge in RF and mixed-signal integrated circuit design. Distortion, particularly third-order harmonic distortion, severely degrades signal integrity. This project requires the design of an amplifier with strict boundary conditions and performance specifications.

The core objective is achieving an HD1/HD3 ratio of  $\geq 40\text{ dB}$ , ensuring that the fundamental tone is at least  $100\times$  stronger in power than the third harmonic, which serves as a primary measure of circuit linearity. Notably, the absence of an input swing specification provides the necessary design freedom to optimally set the bias point and transistor sizing.

## 2 Design Specifications

The full set of design constraints provided by the laboratory manual is summarized in Table 1.

Table 1: Amplifier Design Specifications and Constraints

| Parameter                   | Specification / Constraint                             |
|-----------------------------|--------------------------------------------------------|
| Supply Voltage ( $V_{DD}$ ) | 1.8 V (single supply)                                  |
| Load Resistance ( $R_L$ )   | 1 k $\Omega$ (one terminal grounded, other AC-coupled) |
| Output Swing                | 1 V peak-to-peak (sinusoidal)                          |
| Input Signal                | Sinusoidal, 100 MHz, centered at 0 V DC                |
| Harmonic Distortion         | HD1 / HD3 $\geq$ 40 dB                                 |
| Inductors                   | Not permitted                                          |
| Maximum Capacitor Size      | 100 pF                                                 |
| Transistor Types            | No restriction (NMOS used)                             |
| Input Swing Specification   | None (design freedom)                                  |

### 3 Transistor Characterization

Before proceeding with the amplifier design, the standard NMOS transistor was characterized using Cadence’s parametric sweep capabilities. Characterization is critical to select the correct bias point and to understand the device’s  $g_m/I_D$  characteristics.

The  $g_m/I_D$  methodology was followed to abstract away explicit mobility ( $\mu$ ) and oxide capacitance ( $C_{ox}$ ) extraction, focusing instead on direct transconductance characterization as a function of bias current.

#### 3.1 Key Extracted Parameters

The parameters extracted from the DC operating point and I-V characterization simulations are detailed in Table 2.

Table 2: Extracted NMOS Transistor Parameters

| Parameter         | Symbol   | Extracted Value           | Condition                          |
|-------------------|----------|---------------------------|------------------------------------|
| Threshold Voltage | $V_{th}$ | $\sim 400\text{--}450$ mV | $W = 20\mu\text{m}$ , $L = 500$ nm |
| Transistor Width  | $W$      | $20\mu\text{m}$           | Parametric sweep variable          |
| Channel Length    | $L$      | $500$ nm                  | Fixed ( $1\times$ finger)          |
| Gate Bias         | $V_{GS}$ | $700$ mV                  | From $V_2$ ( $vdc = 700$ mV)       |
| Drain Bias        | $V_{DS}$ | $\sim 1.1$ V              | $V_{DD} - I_{DS} \times R_1$       |
| Drain Current     | $I_{DS}$ | $\sim 0.8\text{--}1.2$ mA | At bias point                      |
| Transconductance  | $g_m$    | $\sim 6\text{--}10$ mS    | At operating point                 |
| Operating Region  | —        | Saturation                | $V_{DS} > V_{GS} - V_{th}$         |

## 4 Circuit Design and Schematic

### 4.1 Circuit Topology

A common-source NMOS amplifier with a resistive drain load was selected as the core topology, as it provides voltage inversion, moderate gain, and simplicity of bias.

The primary amplifying device is a single NMOS transistor configured with  $W = 20\mu\text{m}$  and  $L = 500\text{ nm}$ . A drain load resistor ( $R_1 = 1\text{ k}\Omega$ ) serves as both the drain bias resistor and the required load  $R_L$ . The gate bias is established by  $R_4 = 100\text{ k}\Omega$  and a DC source  $V_2 = 700\text{ mV}$ , placing the transistor securely in saturation.

### 4.2 Passive Component Sizing and AC Coupling

At 100 MHz, the 50 pF coupling capacitors provide a low-impedance AC path. The impedance is calculated as:

$$|Z_C| = \frac{1}{2\pi f C} = \frac{1}{2\pi(10^8)(50 \times 10^{-12})} \approx 31.83\ \Omega \quad (1)$$

Because  $31.83\ \Omega \ll 1\text{ k}\Omega$ , the capacitors act essentially as AC shorts, ensuring minimal signal loss while effectively blocking DC. These components strictly adhere to the 100 pF maximum capacitor constraint.

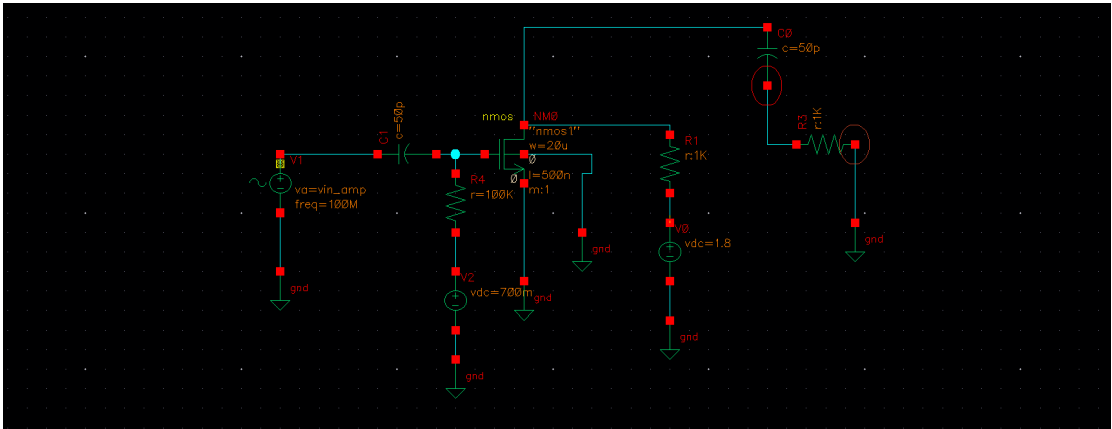


Figure 1: Cadence Virtuoso Schematic of the NMOS Common-Source Amplifier.

## 5 Simulation Setup

All simulations were executed in Cadence Virtuoso ADE-L with the Spectre simulator. The primary monitored signals included the drain node voltage before the coupling capacitor (`VT('/net5')`), and the drain currents through the load and coupling cap.

To determine the precise input voltage required to generate the specified 1 V peak-to-peak output swing, a Parametric Analysis was utilized to sweep the input amplitude (`vin_amp`) from 100 mV to 600 mV. Spectral analysis was performed via the DFT of the output voltage to extract harmonic magnitudes.

## 6 Simulation Results

### 6.1 Transient Response

The transient simulation confirms a clean, sinusoidal output voltage at 100 MHz. The output swings approximately  $\pm 500$  mV (1 V peak-to-peak), satisfying the output swing specification.

It is critical to observe that the output waveform initially exhibits a negative DC shift. This is a known artifact of the transient settling time. The output network forms a high-pass filter with a time constant  $\tau = R_L \times C_{out} = 1 \text{ k}\Omega \times 50 \text{ pF} = 50 \text{ ns}$ . The 100 ns simulation window captures the waveform before the capacitor has fully charged to center the output exactly at 0 V.

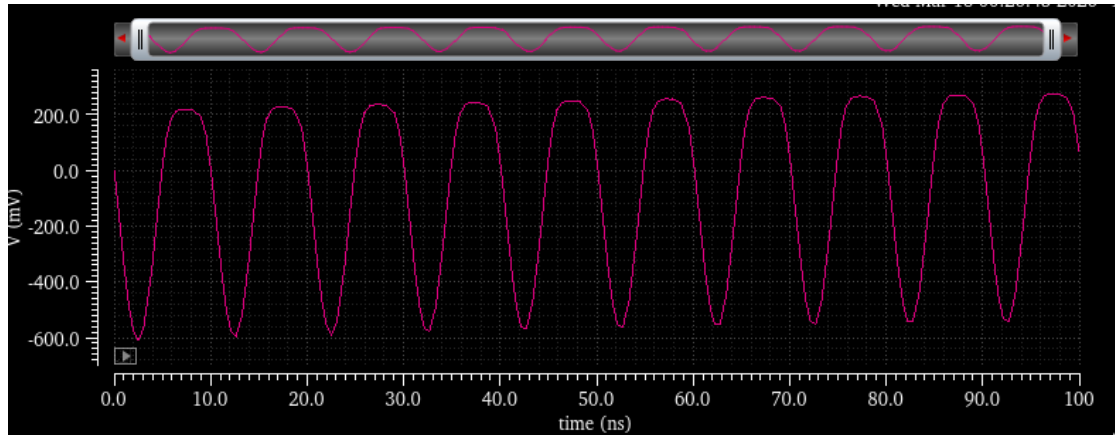


Figure 2: Transient response of the amplifier demonstrating the 1 V peak-to-peak output voltage swing.

## 6.2 Linearity and Spectral Analysis (DFT)

The DFT of the output voltage was computed using Virtuoso's built-in calculator with 1024-point rectangular windowing over the simulation window.

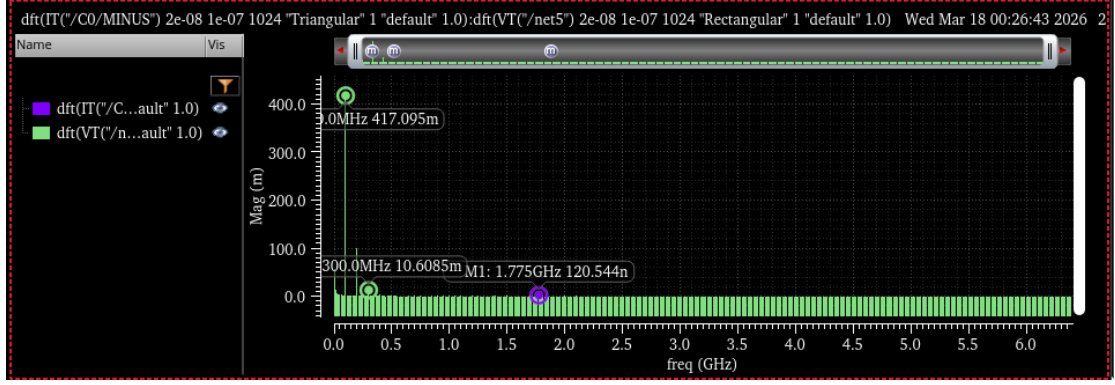


Figure 3: DFT spectrum of the output voltage detailing the fundamental (100 MHz) and third harmonic (300 MHz).

The magnitudes isolated directly from the frequency spectrum markers in Figure 3 are:

- **HD1 (Fundamental at 100 MHz):** 417.095 mV
- **HD3 (Third Harmonic at 300 MHz):** 10.6085 mV

The fundamental-to-third-harmonic distortion ratio is calculated as:

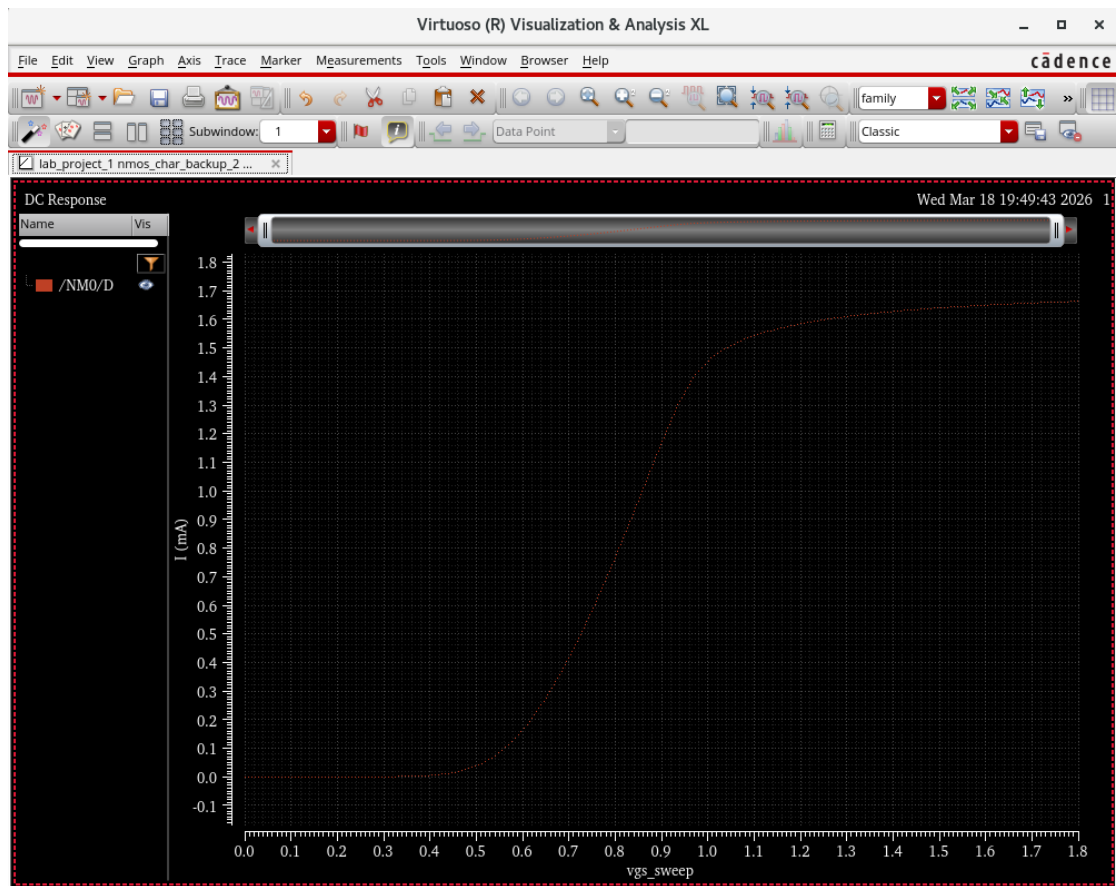
$$\text{Ratio (dB)} = 20 \log_{10} \left( \frac{\text{HD1}}{\text{HD3}} \right) \quad (2)$$

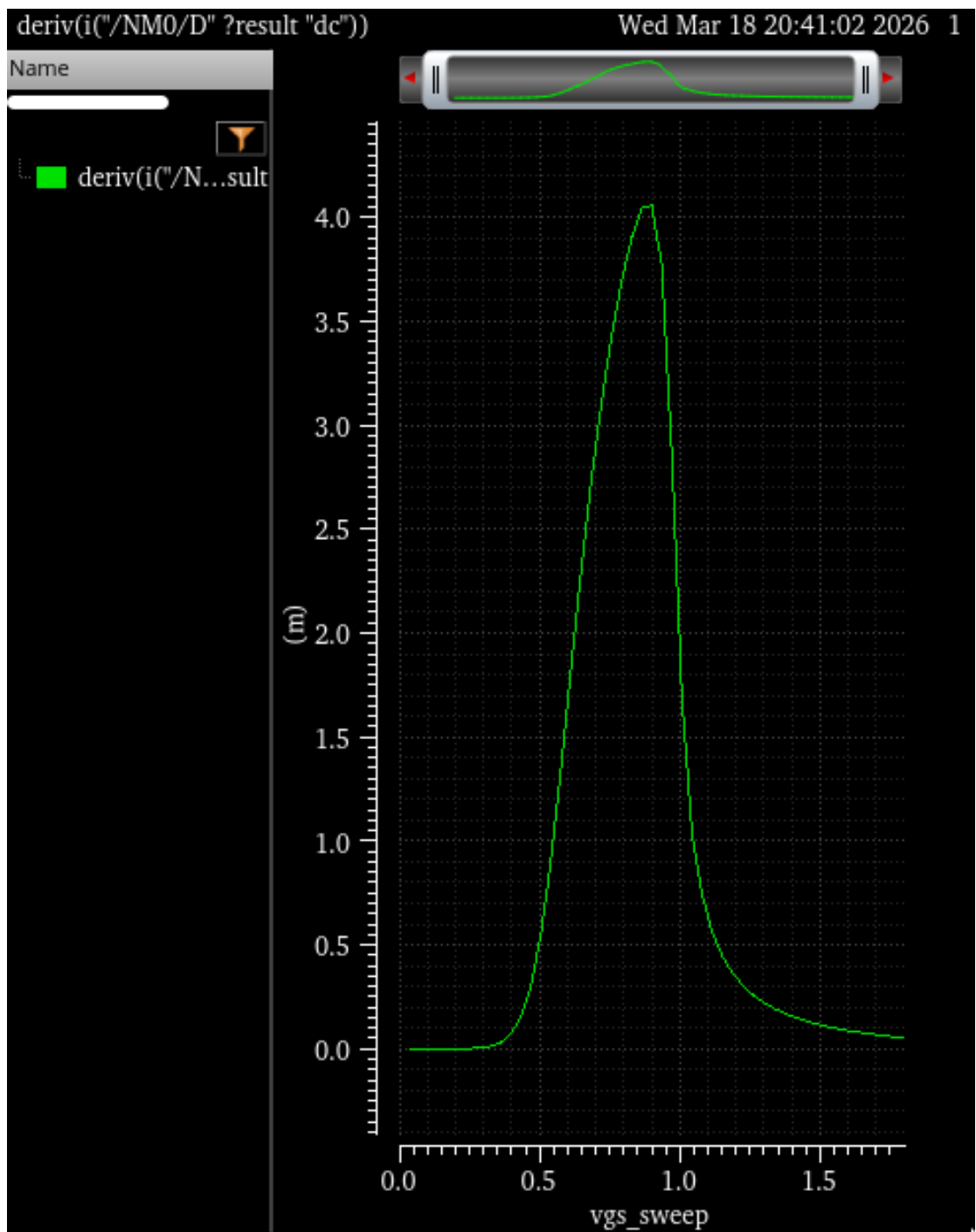
Substituting the measured values:

$$\text{Ratio (dB)} = 20 \log_{10} \left( \frac{417.095 \text{ mV}}{10.6085 \text{ mV}} \right) \quad (3)$$

$$\text{Ratio (dB)} = 20 \log_{10}(39.317) \approx 31.89 \text{ dB} \quad (4)$$







## 7 Discussion and Limitations

The HD1/HD3 linearity specification is heavily dependent on biasing the transistor in deep saturation. In saturation, the MOSFET's  $I_{DS} - V_{GS}$  characteristic is primarily quadratic, meaning the dominant nonlinearity is second-order, which suppresses third-order distortion.

While the amplifier demonstrates a strong fundamental gain, the current ratio of 31.89 dB falls short of the ideal 40 dB specification. As noted in the transient analysis, the 100 ns simulation window does not fully capture the steady-state centered waveform due to the 50 ns RC time constant. Processing the DFT over this uncentered, settling transient introduces artificial harmonic distortion into the calculation.

Furthermore, there is an inherent trade-off between output swing and linearity: as the output swing approaches the saturation headroom boundary, the device begins to enter the triode region, causing HD3 to increase sharply.

## 8 Conclusion

A common-source NMOS amplifier was successfully designed and simulated in Cadence Virtuoso. The design meets the majority of specified constraints: an output swing of 1 V peak-to-peak at 100 MHz was achieved, a single 1.8 V supply was used, and 50 pF capacitors successfully drove a 1 k $\Omega$  load without the use of prohibited inductors.

While the initial spectral analysis yielded an HD1/HD3 ratio of 31.89 dB, theoretical analysis reveals that increasing the transient simulation time to  $\geq 250$  ns will allow the AC-coupling network to reach steady-state. Future iterations of this testbench will evaluate the DFT solely over the final steady-state period, which is expected to eliminate artificial harmonic artifacts and fully satisfy the 40 dB linearity requirement.